A PROPERTY LOCAL ON THE TARGET IS PRESERVED BY FLAT, SYNTOMIC, ETALE AND OPEN BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Localness of a property of morphisms on the target for a topology. Same chapter, the definition of tau localness on the target, named in the statement at lines 5555-5556. Supplies the axiom that transfers the property along a covering family.
- A single flat (or fppf, syntomic, etale, open immersion) morphism forms a covering family. Standard, used at proof lines 5564-5565. Supplies the covering family into which the given morphism is fitted.

Lemma 0.1. *Let $\tau \in \{fpqc, fppf, syntomic, smooth, étale, Zariski\}$. Let $\mathcal{P}$ be a property of morphisms which is τ local on the target. Let $f : X \to Y$ have property $\mathcal{P}$. For any morphism $Y' \to Y$ which is flat, resp. flat and locally of finite presentation, resp. syntomic, resp. étale, resp. an open immersion, the base change $f' : Y' \times_Y X \to Y'$ of f has property $\mathcal{P}$.*

Proof. This is true because we can fit $Y' \to Y$ into a family of morphisms which forms a τ-covering. □

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